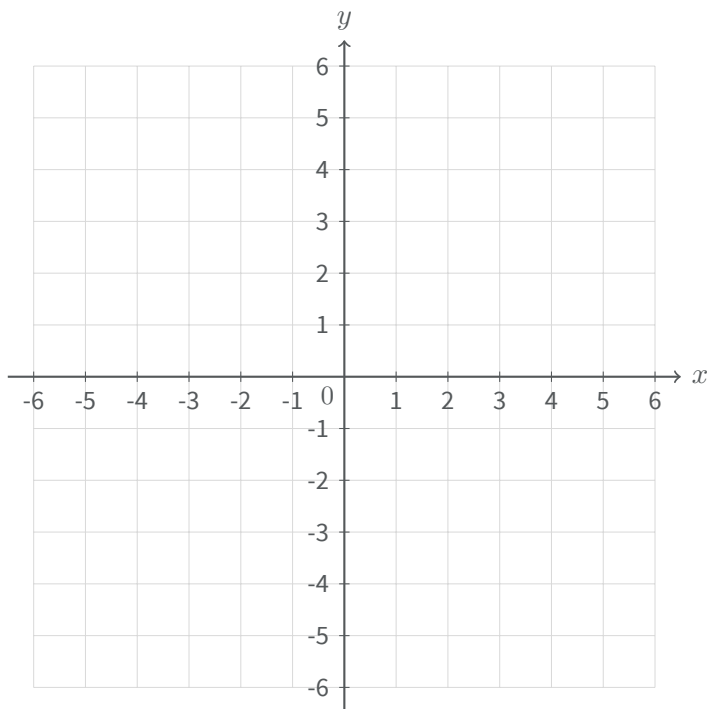


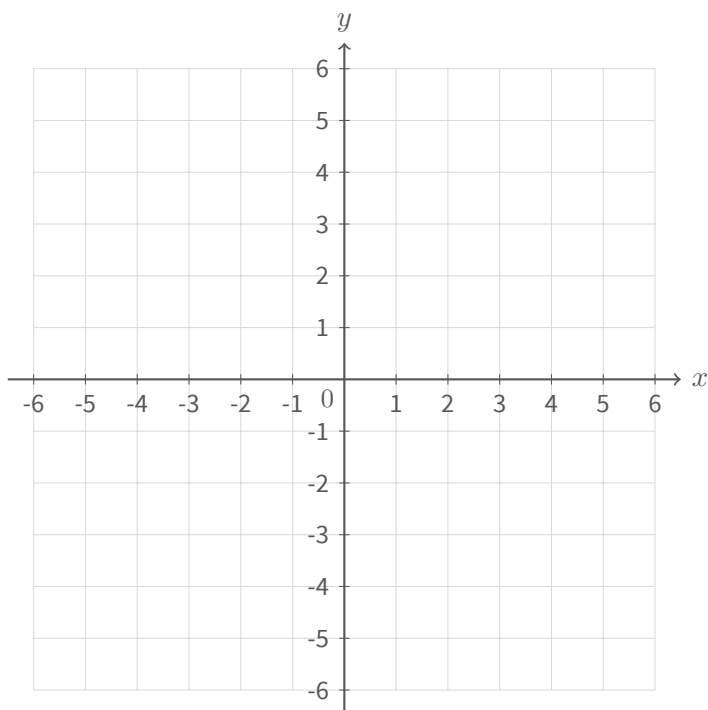
M^5 Day 7: Graphing Quadratics Name: _____

1. Sketch the following parabolas, labeling x and y -intercepts:

a. $y = -x^2 + x + 2$



b. $y = -x + 3 + 2x^2$



2. Give an example of a parabola with each of the following properties:

a. The parabola opens down and has x -intercepts at -2 and 1 .

b. The parabola has a maximum at $x = 1$ and a y -intercept of -2 .

c. The parabola has no real roots and has a minimum value.

d. The parabola has exactly one real root.

M^5 Day 7: Function notation

Name: _____

1. Evaluate the function at the indicated values:

a. $f(x) = x^2 + 2x$ at $f(0)$, $f(3)$, $f(-x)$, $f(1/a)$.

b. $f(x) = |x|/x$ at $f(-2)$, $f(-1)$, $f(5)$, $f(x^2)$.

c. $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ x + 1 & \text{otherwise} \end{cases}$ at $f(-2)$, $f(-1)$, $f(0)$, $f(1)$.

d. $g(t) = t^2 + 1$ at $g(x + 2)$ and $g(x) + g(2)$.

e. $h(z) = \sqrt{1 + z^2}$ at $h(1)$ and $h(-1)$.